

On Nonuniform Sampling of Bandlimited Signals Associated with the Fractional Fourier Transform

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Abstract

The uniform sampling theorem and the reconstruction formulae associated with the fractional Fourier transform (FrFT) have been deduced in the literature, but the nonuniform sampling is yet to receive attention in the fractional Fourier domain. This paper focus on a special kind of non-uniform sampling process associated with the fractional Fourier transform. A nonuniform sampling model is first introduced and then, based on a useful lemma, the reconstruction formula for the nonuniform sampled signal points associated with the fractional Fourier transform is deduced. In addition, the classical results in the Fourier domain can be seen as a special case of our work.

1. Introduction

The fractional Fourier transform (FRFT) [1], [2],[16], [9], [10], [13], [20], has in recent time been one of the most frequently used tool in the signal processing [2], [10], [18] community and other scientific discipline. Its application has seen various developments in the time-varying or nonstationary signal processing [19], fiber optics [6],[7], bulk optics [4], [5],[8], filters, radars and time-frequency analysis.

The FrFT has also extended its application to one of the most fundamental works of digital signal processing - the sampling process. The extension of the classical Shannon sampling theorem to the fractional Fourier domain (FFD) has also received attention in the literature. In [24], the authors presented the possibility of using the FRFT for the processing of nonstationary signals at no extra cost anywhere the traditional FFT schemes have been inefficient. There is therefore a potential for generalization and improvement, using the FrFT wherever the FT and the frequency domain concepts have been employed.

Uniform sampling of band-limited signals has been treated fairly in both the FFD and the traditional Fourier domain whereas nonuniform sampling has received attention only in the Fourier domain. Nonuniform sampling measurements do occur naturally in several applications, including optics, tomography, frequency modulation, phase, delta, or pulse position demodulation etc. Data, such as astronomical or geophysical, are most often not available for measurement at certain points of the domain and hence impossible to sample uniformly. There are also, several factors like jitter, clipping, lost or delayed packets in packet-oriented telecommunication systems etc. which can accidentally lead to irregularly distributed sets of samples. Theorems have been proposed to address the above mentioned but in the traditional Fourier domain [25,26]. Compared to the Fourier transform, the FRFT is the most suitable tool for the processing of nonstationary signals. The FRFT can also be seen as the extension of the Fourier transform. It is therefore worthwhile to deduce the non-uniform sampling formulae for band-limited signals in the FFD.

In this paper we seek to address the issue concerned with the reconstruction of an original signal from a special kind of nonuniform sampled band-limited signal in the FFD. The paper is organized as follows: In section 2, we introduce the FRFT concept and the sampling theorem. In section 3, the main results of our work covering the nonuniform sampling and special cases have been presented and the conclusion of the paper in section 4.

2. The FRFT and the sampling theorem

2.1. Continuous time Fractional Fourier Transform

Recall [2] that the continuous-time FRFT of an analog signal $f(t)$ with angle α can be defined as

$$X_{\alpha}(u) = \int_{-\infty}^{\infty} K_{\alpha}(u, t) f(t) dt \quad (1)$$

where $K_{\alpha}(u, t)$ is the kernel function of the FRFT:

$$K_{\alpha}(u, t) = \begin{cases} A_{\alpha} \exp(j\frac{t^2}{2} \cot \alpha - jut \csc \alpha + j\frac{u^2}{2} \cot \alpha), & \alpha \neq k\pi \\ \delta(t-u), & \alpha = 2k\pi \\ \delta(t+u), & \alpha = \pi + 2k\pi \end{cases}, A_{\alpha} = \sqrt{\frac{1-j\cot \alpha}{2\pi}} \quad (2)$$

The inverse FRFT (IFRFT) can therefore be expressed as:

$$f(t) = \int_{-\infty}^{\infty} X_{\alpha}(u) K_{-\alpha}(t, u) du \quad (3)$$

The u domain is usually called the fractional Fourier domain; the time and frequency domain can be considered as its special cases when $\alpha = 0$ and $\alpha = \pi/2$ respectively. For a deeper understanding of the FRFT, the reader is referred to [3],[10],[11],[12] as well as our extensive list of references.

Definition 1. A signal $f(t)$ is said to be Ω_{α} band-limited in the Fractional Fourier domain if

$$F_{\alpha}(u) = 0 \quad \text{for } |u| > \Omega_{\alpha}$$

where, Ω_{α} is the bandwidth of the signal $f(t)$ in the Fractional Fourier domain.

Lemma 1. Let $f(t)$ be a band-limited signal in the Fractional Fourier domain such that $F_{\alpha}(u) = 0$ for $|u| > \Omega_{\alpha}$, and

$$g(t) = \int_{-\Omega_{\alpha}}^{\Omega_{\alpha}} F_{\alpha}(u) e^{-j\frac{u^2}{2} \cot \alpha + jut \csc \alpha} du$$

then $g(t)$ is $\Omega_{\alpha} \csc \alpha$ band-limited signal in the conventional Fourier transform domain.

Proof: see [7]

Lemma 2. Let $f(t)$ be a band-limited signal in the fractional Fourier domain with $F_{\alpha}(u) = 0$ for $|u| > \Omega_{\alpha}$, then it can be reconstructed exactly from its samples, $f(nT)$, $n = 0, \pm 1, \pm 2, \dots$, if the sampling period T satisfy:

$$\frac{\pi}{T} > \Omega_{\alpha} \csc \alpha \quad (4)$$

The original signal $f(t)$ can then be reconstructed [6] from the following:

$$f(t) = e^{-j\frac{t^2}{2} \cot \alpha} \sum_{n=-\infty}^{\infty} e^{j\frac{(nT)^2}{2} \cot \alpha} f(nT) \frac{\sin[\Omega(t-nT)\csc \alpha]}{\Omega(t-nT)\csc \alpha} \quad (5)$$

Proof: see [15]

The forward and inverse discrete-time fractional Fourier transform (DTFRFT) [10] of a signal $x(n)$, can be represented as

$$X_{\alpha}(u) = \sqrt{\frac{1-j\cot \alpha}{2\pi}} e^{j\frac{u^2}{2} \cot \alpha} \sum_n e^{j\frac{n^2 T^2}{2} \cot \alpha - jn u T \csc \alpha} x(n) \quad (6)$$

$$x(n) = T \sqrt{\frac{1+j\cot \alpha}{2\pi}} \int_{\langle u^{\alpha} \rangle} X_{\alpha}(u) e^{-j\frac{n^2 T^2}{2} \cot \alpha + jn u T \csc \alpha} du \quad (7)$$

where T is the sampling period and $\langle u^{\alpha} \rangle$ the integral interval of size $\frac{2\pi \sin \alpha}{T}$ taken

$$\text{as } \left[-\frac{\pi \sin \alpha}{T}, \frac{\pi \sin \alpha}{T} \right]$$

2.2. The discrete time FRFT and the sampling theorem

An important operation required in various application domains, such as communication and controls, is the sampling of the continuous-time signal (for instance $x(t)$). In uniform sampling, the sample values of $x(t)$ are equal to the values $x(nT)$ where n is the integer index $n = 0, \pm 1, \pm 2, \dots$ and T the sampling interval. To simplify the analysis of the sampling operation, the sampled version of $x(t)$ is often expressed in the form $x(t)s(t)$ where $s(t)$ the impulse train can be defined as

$$s(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT)$$

The output of an ideal sampler that instantaneously measures an analog signal $x(t)$ at the sampling instants $t = nT$ [22] can be expressed as

$$\begin{aligned} \hat{x}(t) &= x(t) s(t) \\ &= x(t) \sum_{n=-\infty}^{+\infty} \delta(t - nT) \end{aligned} \quad (8)$$

where T is the sampling interval. Applying the product theorem [23] the discrete-time FRFT of $\hat{x}(t)$ can be written as

$$S_{\alpha}(u) = \frac{1}{T} e^{j\frac{u^2}{2} \cot \alpha} \left[X_{\alpha}(u) e^{-j\frac{u^2}{2} \cot \alpha} * \sum_{n=-\infty}^{+\infty} \delta\left(u - n \frac{2\pi \sin \alpha}{T}\right) \right] \quad (9a)$$

$$= \frac{1}{T} e^{\frac{j\pi^2}{2} \cot \alpha} \left\{ \sum_{n=-\infty}^{+\infty} X_{\alpha} \left(u - \frac{2n\pi \sin \alpha}{T} \right) e^{-\frac{j}{2} \left(u - \frac{2n\pi \sin \alpha}{T} \right)^2 \cot \alpha} \right\} \quad (9b)$$

where $X_{\alpha}(u)$ is the continuous-time FRFT of the analog signal $x(t)$ and n the harmonic order.

3. Nonuniform Sampling

3.1. Nonuniform sampling model

One of the generalized non-uniform sample point distribution is obtained from the migration of a finite number of sample points in a uniform distribution [17]. For this kind of nonuniform sampling pattern, Yen [17] deduced the classical reconstruction theorem of the original signal in the traditional Fourier domain. His method is only suitable for band-limited signal in the traditional Fourier domain, and may not be suitable for non-band-limited signal in the Fourier domain. This nonuniform sampling model is as shown in Fig.1. It shows N uniform sample points located at $t = n_p/2W$, where $n_p (p=1, 2, \dots, N)$ are N distinct integers, have been migrated to N new positions $t = t_p$, where $2Wt_p$ is not an integer.

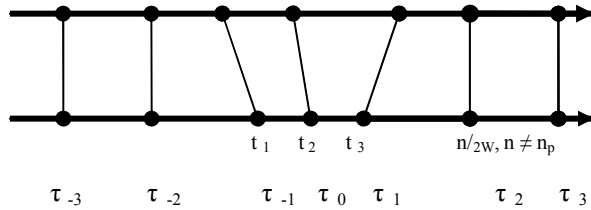


Fig. 1 Nonuniform sampling model.

Based on the above knowledge, we focus on this kind of nonuniform sampling model to obtain the reconstruction theorem for the original signal in the fractional Fourier domain in the next section.

3.2. The nonuniform sampling theorem

Theorem 1: If $f(t)$ is a band-limited signal in the fractional Fourier domain such that $F_{\alpha}(u) = 0$ for $|u| > \Omega_{\alpha}$; and N uniform sample points located at $t = n_p/2\Omega_{\alpha}$ are moved to N new positions $t = t_p$ where n_p with $p=1, 2, \dots, N$ are N distinct integers, then, the original signal $f(t)$ can be

reconstructed from the nonuniform sampled points τ_m and the interpolation formula can be written as follows

$$f(t) = e^{-j\frac{t^2}{2} \cot \alpha} \sum_{m=-\infty}^{\infty} f(\tau_m) e^{j\frac{\tau_m^2}{2} \cot \alpha} \psi_m(t) \quad (10)$$

where

$$\psi_m(t) = \begin{cases} \frac{\prod_{q=1}^N (t - t_q) \prod_{q=1}^N \left(\frac{n}{2\Omega_{\alpha} \csc \alpha} - \frac{n_q}{2\Omega_{\alpha} \csc \alpha} \right)}{\prod_{q=1}^N \left(t - \frac{n_q}{2\Omega_{\alpha} \csc \alpha} \right) \prod_{q=1}^N \left(\frac{n}{2\Omega_{\alpha} \csc \alpha} - t_q \right)} \\ \times \frac{(-1)^n \sin[(2\pi\Omega_{\alpha} \csc \alpha)t]}{\pi[(2\Omega_{\alpha} \csc \alpha)t - n]}, \tau_m = \frac{n}{2\Omega_{\alpha} \csc \alpha} \neq \frac{n_q}{2\Omega_{\alpha} \csc \alpha}, \\ \frac{\prod_{q=1}^N (t - t_q) \prod_{q=1}^N \left(t_p - \frac{n_q}{2\Omega_{\alpha} \csc \alpha} \right)}{\prod_{q=1}^N \left(t - \frac{n_q}{2\Omega_{\alpha} \csc \alpha} \right) \prod_{q=1}^N (t_p - t_q)} \cdot \frac{\sin[(2\pi\Omega_{\alpha} \csc \alpha)t]}{\sin[(2\pi\Omega_{\alpha} \csc \alpha)t_p]}, \tau_m = t_p \end{cases} \quad (11)$$

Proof: Since $f(t)$ is a band-limited signal in the fractional Fourier domain, if we let

$$g(t) = \int_{-\Omega_{\alpha}}^{\Omega_{\alpha}} F_{\alpha}(u) e^{-j\frac{u^2}{2} \cot \alpha + j u t \csc \alpha} du,$$

then from lemma 1, $g(t)$ is a $\Omega_{\alpha} \csc \alpha$ band-limited signal in the conventional Fourier transform domain. Applying the classical results for the nonuniform sampling theory [17] to the signal $g(t)$, results in

$$g(t) = \sum_{m=-\infty}^{\infty} g(\tau_m) \psi_m(t) \quad (12)$$

The relationship between the $f(t)$ and $g(t)$ can therefore be written as:

$$g(t) = f(t) \sqrt{\frac{2\pi}{1 + j \cot \alpha}} e^{j\frac{t^2}{2} \cot \alpha} \quad (13)$$

From expressions (12) and (13), the final results can be deduced as (10).

3.3. Special Cases

1) When $\alpha = \pi/2$, equation (10) reduces to

$$f(t) = \sum_{m=-\infty}^{\infty} f(\tau_m) \psi_m(t) \quad (14)$$

Equation (14) is the classical result in the Fourier domain proposed by Yen [17]

2) When one sample uniformly at $\tau_m = mT$, expression (10) reduces to

$$f(t) = e^{-j\frac{t^2}{2} \cot \alpha} \sum_{m=-\infty}^{\infty} f(mT) e^{j\frac{(mT)^2}{2} \cot \alpha} \psi_m(t) \quad (15)$$

Expression (15) is the well known sampling theorem of Xia [15] in the fractional Fourier domain.

- 3) When $\alpha = \pi/2$ and $\tau_m = mT$, expression (10) reduces to (16), which is the Shannon's classical sampling theorem

$$f(t) = \sum_{m=-\infty}^{\infty} f(mT) \psi_m(t) \quad (16)$$

From above, the well known results in the Fourier domain can be seen as special cases of the proposed theory, furthermore our result can be used for nonstationary signal processing - such as chirp signal processing.

4. Conclusion

Based on the properties of the FRFT and the uniform sampling theorem in the fractional Fourier domain, a nonuniform sampling model encountered in practical applications is introduced and the reconstruction condition and formula from this kind of nonuniform sampling model in the fractional Fourier domain is derived. In addition, the new results can also be seen as a generalization of the classical results from the Fourier domain to the fractional Fourier domain. Our future work is to extend it into the multidimensional domain and apply the real results in the signal processing society. To prove the effectiveness of our result, we are currently pursuing practical verification of our result using chirp signal. This will be published in a separate paper.

5. References

- [1] L. B. Almeida, "An introduction to the angular Fourier transform", in *Proc. IEEE Int. conf. Acoust., Speech, Signal Processing* (Minneapolis, MN), April 1993.
- [2] L. B. Almeida, "The fractional Fourier transform and time-frequency representations", *IEEE Trans. Signal Processing*, vol. 42, pp. 3084-3091, Nov. 1994.
- [3] H. M. Ozaktas, B. Barshan, D. Mendlovic and L. Onural, "Convolution, filtering and multiplexing in fractional Fourier domains and their relation to chirp and wavelet transform", *J. Opt. Soc. Am. A.*, vol. 11, 1994, pp. 547-559.
- [4] L. M. Bernado, and O. D. D. Soares, "Fractional Fourier transforms and imaging", *J. Opt. Soc. Am. A.*, vol. 11, no. 10, pp. 2622-2626, Oct. 1994.
- [5] A. W. Lohmann, "Image rotation, Wigner rotation and the fractional Fourier transform", *J. Opt. Soc. Am. A.*, vol. 10, no. 10, pp. 2181-2186, Oct. 1993.
- [6] D. Mendlovic, and H. M. Ozaktas, "Fractional Fourier transforms and their optical implementation: I", *J. Opt. Soc. Am. A.*, vol. 10, no. 9, pp. 1875-1881, Sept. 1993.
- [7] H. M. Ozaktas, and D. Mendlovic, "Fractional Fourier transforms and their optical implementation: II", *J. Opt. Soc. Am. A.*, vol. 10, no.12, pp. 2522-2531, Dec. 1993.
- [8] H. M. Ozaktas, and D. Mendlovic, "Fractional Fourier optics", *J. Opt. Soc. Am. A.*, vol. 12, no.4, pp. 743-751, Apr. 1995.
- [9] N. Wiener, "Herrmitian polynomials and Fourier analysis", *J. Math. Phys. MIT*, vol. 18, pp. 70-73, 1929.
- [10] H. M. Ozaktas, M. A. Kutay, and D. Mendlovic, "Introduction to the fractional Fourier transform and its applications. P. W. Hawkes, ed., *Advances in Imaging and Electron Physics*, vol.106, Academic Press, San Diego, California, 1999, pp. 239-291.
- [11] Ran Tao, Lin Qi, and Yue Wang, *Theory and Applications of the Fractional Fourier Transform*. Beijing: *Tsinghua Univ. Press*, 2004.
- [12] H. M. Ozaktas, Z. Zalevsky, and M. A. Kutay, *The fractional Fourier transform with Applications in Optics and Signal Processing*, John Wiley & Sons, 2001.
- [13] A. W. Lohmann, D. Mendlovic, and Z. Zalevsky. *Fractional transformations in optics. Progress in Optics*, New York: Elsevier, vol. 38, ch. 4, 1998.
- [14] A. Papoulis, *Signal Analysis*, New York: McGraw-Hill, 1977
- [15] X. Xia, "On Bandlimited signals with fractional Fourier transform", *IEEE Processing Lett.*, vol.3, no. 3, pp.72-74, Mar.1996.
- [16] V.Namias, "The fractional order Fourier transform and its application to quantum mechanics", *J. Inst. Math.Appl.*, vol.25,pp.241-265,1980.
- [17] J. L. Yen, "On nonuniform sampling of bandwidth-limited signals", *IEEE Trans. Circuit Theory*, vol. CT-3, pp. 251-257, Dec. 1956
- [18] A. I. Zayed, "A class of Fractional Integer transforms: A generalization of the fractional Fourier transform", *IEEE Trans. Signal Processing*, vol. 50, no. 3, Mar. 2002.
- [19] M. A. Kutay, H. M. Ozaktas, and O. Arikan, "Optimal filtering in fractional Fourier domains", *IEEE Trans. Signal Processing*, vol. 45, no. 5, pp. 1129-1143, May 1997.
- [20] A. I. Zayed, "On the Relationship Between the Fourier and Fractional Fourier Transform", *IEEE Signal Processing Lett.*, vol. 3, pp. 310-311, Dec. 1996.
- [21] T. Erseghe, P. kraniauskas and G. Cariolaro, "Unified fractional Fourier transform and sampling theorem", *IEEE Trans. Signal Processing*, vol. 47, pp 3419-4323, Dec. 1999.
- [22] R. E. Crochiere and L. R. Rabiner, "Interpolation and decimation of digital signals - a tutorial review", in *Proc. IEEE*, vol. 69, no. 3, pp. 300, Mar. 1981.
- [23] L. B Almeida, "Product and convolution theorems for the fractional Fourier Transform", *IEEE Signal Processing Lett.*, vol. 4, pp. 15-17, Jan. 1997.
- [24] H. M. Ozaktas, O. Arikan, M. A. Kutay and G. Bozdagi, "Digital computation of the fractional Fourier transforms," *IEEE Trans. Signal Processing*, vol. 44, Sept. 1996, pp. 2141-2150.
- [25] M.R. Belmont, "Nonuniform sampling specifically for finite-length data," *IEE Proceedings-F*, vol. 140, no. I, pp. 55-62, Feb. 1993.
- [26] T.Schanze, "Sinc interpolation of discrete periodic signals," *IEEE Trans. Signal Processing*, vol. 43, pp. 1502-1503, Jun.1995.